

LungTIME-C01 Numeric Integrity Audit

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1 Scope

This report summarizes numeric-integrity checks aligned with Heathers' data-technique family:

- internal rounding consistency for count-percent table cells

- scrutiny GRIM / GRIMMER / DEBIT and duplicate-value checks
- rounding-bias diagnostics for reported decimal patterns
- statistical-consistency checks via `statcheck`

2 Load Outputs

```
repo_root <- normalizePath("../", winslash = "/", mustWork = TRUE)
report_dir <- file.path(repo_root, "reports", "numeric", "lungtime")

required <- c(
  "numeric_row_results.csv",
  "numeric_summary.csv",
  "numeric_package_status.csv",
  "numeric_scrutiny_grim_raw.csv",
  "numeric_scrutiny_grim_audit.csv",
  "numeric_scrutiny_grimmer_raw.csv",
  "numeric_scrutiny_grimmer_audit.csv",
  "numeric_scrutiny_debit_raw.csv",
  "numeric_scrutiny_debit_audit.csv",
  "numeric_scrutiny_duplicates.csv",
  "numeric_scrutiny_rounding_bias.csv",
  "numeric_statcheck_raw.csv",
  "numeric_standardized_results.csv"
)
missing <- required[!file.exists(file.path(report_dir, required))]
if (length(missing) > 0) {
  stop("Missing files: ", paste(missing, collapse = ", "))
}

row_results <- read.csv(file.path(report_dir, "numeric_row_results.csv"), check.names = FALSE)
summary_tbl <- read.csv(file.path(report_dir, "numeric_summary.csv"), check.names = FALSE)
pkg_status <- read.csv(file.path(report_dir, "numeric_package_status.csv"), check.names = FALSE)
grim_raw <- read.csv(file.path(report_dir, "numeric_scrutiny_grim_raw.csv"), check.names = FALSE)
grim_audit <- read.csv(file.path(report_dir, "numeric_scrutiny_grim_audit.csv"), check.names = FALSE)
grimmer_raw <- read.csv(file.path(report_dir, "numeric_scrutiny_grimmer_raw.csv"), check.names = FALSE)
grimmer_audit <- read.csv(file.path(report_dir, "numeric_scrutiny_grimmer_audit.csv"), check.names = FALSE)
debit_raw <- read.csv(file.path(report_dir, "numeric_scrutiny_debit_raw.csv"), check.names = FALSE)
debit_audit <- read.csv(file.path(report_dir, "numeric_scrutiny_debit_audit.csv"), check.names = FALSE)
duplicates_raw <- read.csv(file.path(report_dir, "numeric_scrutiny_duplicates.csv"), check.names = FALSE)
rounding_bias <- read.csv(file.path(report_dir, "numeric_scrutiny_rounding_bias.csv"), check.names = FALSE)
```

```

statcheck_raw <- read.csv(file.path(report_dir, "numeric_statcheck_raw.csv"), check.names = FALSE)
standardized <- read.csv(file.path(report_dir, "numeric_standardized_results.csv"), check.names = FALSE)

compact_character <- function(x, width = 48) {
  x <- as.character(x)
  x[is.na(x)] <- ""
  too_long <- nchar(x) > width
  x[too_long] <- paste0(substr(x[too_long], 1, width - 3), "...")
  x
}

compact_df <- function(df, cols = NULL, n = 20, char_width = 48) {
  out <- df
  if (!is.null(cols)) {
    keep <- intersect(cols, names(out))
    out <- out[, keep, drop = FALSE]
  }
  if (nrow(out) > n) {
    out <- utils::head(out, n)
  }
  is_char_like <- vapply(out, function(col) is.character(col) || is.factor(col), logical(1))
  out[is_char_like] <- lapply(out[is_char_like], compact_character, width = char_width)
  out
}

render_table <- function(df, note, cols = NULL, n = 20, digits = 4, char_width = 48) {
  if (nrow(df) > 0) {
    knitr::kable(compact_df(df, cols = cols, n = n, char_width = char_width), digits = digits)
  } else {
    cat(note)
  }
}

```

3 Section 4 Summary Metrics Key

These are screening indicators from Heathers' data-techniques framework, not proof of misconduct.

- `n_rows`: number of categorical count-percent cells checked for arithmetic consistency.
- `n_rounding_flags`: rows with `|reported_percent - computed_percent| >= 0.2`.

- `rounding_flag_rate`: `n_rounding_flags / n_rows`; larger values suggest widespread reporting drift.
- `median_abs_percent_delta`, `max_abs_percent_delta`: typical and worst discrepancy magnitude.
- `scrutiny_cases_n`: total canonical rows prepared for scrutiny-family methods.
- `grim_cases`, `grim_incons_cases`: testable rows and inconsistency count from `scrutiny::grim_map`.
- `grimmer_cases`, `grimmer_incons_cases`: testable rows and inconsistency count from `scrutiny::grimmer_map`.
- `debit_cases`, `debit_incons_cases`: binary-proportion rows testable by DEBIT and their inconsistency count.
- `duplicate_flag_rows`: rows with at least one duplicated field (`x`, `sd`, or `n`) in duplicate checks.
- `rounding_bias_groups`: (`trial_id`, `digits_x`) groups assessed for directional rounding imbalance.
- `statcheck_cases`, `statcheck_errors`, `statcheck_decision_errors`: parsed test rows, any inconsistencies, and significance-changing inconsistencies.
- `scrutiny_seq_enabled`: whether optional sequence-space checks were executed.

Empty method tables can mean “no eligible rows”, “package unavailable”, or execution failure; inspect `numeric_package_status.csv` execution notes for attribution.

```
if (nrow(summary_tbl) > 0) {
  summary_long <- data.frame(
    metric = names(summary_tbl),
    value = as.character(summary_tbl[1, ]),
    stringsAsFactors = FALSE
  )
  knitr::kable(summary_long)
} else {
  cat("No summary rows available.")
}
```

metric	value
trial_id	lungtime_c01_s41591_025_04181
n_rows	54
n_rounding_flags	2
rounding_flag_rate	0.037037037037037
n_reported_p	54
median_abs_percent_delta	0.03333333333333385
max_abs_percent_delta	0.938095238095236
scrutiny_cases_n	2

metric	value
grim_available	TRUE
grim_cases	2
grim_incons_cases	0
grimmer_available	TRUE
grimmer_cases	0
grimmer_incons_cases	0
debit_available	TRUE
debit_cases	0
debit_incons_cases	0
duplicates_available	TRUE
duplicate_flag_rows	2
rounding_bias_available	TRUE
rounding_bias_groups	1
statcheck_available	TRUE
statcheck_rows	13
statcheck_cases	0
statcheck_errors	0
statcheck_decision_errors	0
rsprite2_rows	27
scrutiny_seq_enabled	FALSE

4 Package Availability

```
render_table(
  pkg_status,
  "No package-status output.",
  cols = c("package", "installed", "version", "objective", "executed", "execution_note"),
  n = 20,
  char_width = 42
)
```

pack- age	in- stalled	ver- sion	objective	exe- cuted	execution_note
scrutiny	TRUE	0.6.1	GRIM/GRIM- MER/DEBIT/duplica- tion/rounding...	TRUE	grim= ok ; grimmer= No GRIMMER-eligible...

pack- age	in- stalled	ver- sion	objective	exe- cuted	execution_note
rsprite2	FALSE		SPRITE-style proportion and distributio...	FALSE	Package execution not yet implemented i...
statcheck	TRUE	1.5.0	Recompute p-values from reported statis...	TRUE	ok

5 GRIM Results

```
render_table(
  grim_audit,
  "No GRIM audit output (package unavailable or no GRIM-eligible rows).",
  n = 20
)
```

in- cons_cases	in- all_cases	in- cons_rate	in- mean_grim_probs	in- cons_to_prob	testable_cases	testable_rate
0	2	0	0	NA	0	0

```
render_table(
  grim_raw,
  "No GRIM row-level output.",
  cols = c("case_id", "source_unit", "x", "n", "consistency", "probability"),
  n = 20
)
```

case_id	source_unit	x	n	consistency	probability
case_0001	Age (year, median, range) [early_tod]	61	105	TRUE	0
case_0002	Age (year, median, range) [late_tod]	60	105	TRUE	0

6 GRIMMER Results

```
render_table(
  grimmer_audit,
  "No GRIMMER audit output (package unavailable or no GRIMMER-eligible rows).",
  n = 20
)
```

No GRIMMER audit output (package unavailable or no GRIMMER-eligible rows).

```
render_table(
  grimmer_raw,
  "No GRIMMER row-level output.",
  cols = c("case_id", "variable", "level", "group", "x", "sd", "n", "consistency", "reason"),
  n = 20,
  char_width = 36
)
```

No GRIMMER row-level output.

7 DEBIT Results

```
render_table(
  debit_audit,
  "No DEBIT audit output (package unavailable or no DEBIT-eligible rows).",
  n = 20
)
```

No DEBIT audit output (package unavailable or no DEBIT-eligible rows).

```
render_table(
  debit_raw,
  "No DEBIT row-level output.",
  cols = c("case_id", "source_unit", "x", "sd", "n", "consistency", "rounding"),
  n = 20,
  char_width = 40
)
```

No DEBIT row-level output.

8 Duplication And Rounding-Bias Results

```
render_table(
  duplicates_raw,
  "No duplicate-check rows available.",
  cols = c("case_id", "variable", "level", "group", "x_dup", "sd_dup", "n_dup", "x_n", "sd_n", "n_n"),
  n = 20,
  char_width = 36
)
```

case_id	variable	level	group	x_dup	sd_dup	n_dup	x_n	sd_n	n_n
case_0001	Age (year, median, range)	all	early_tod	FALSE	TRUE	TRUE	1	2	2
case_0002	Age (year, median, range)	all	late_tod	FALSE	TRUE	TRUE	1	2	2

```
render_table(
  rounding_bias,
  "No rounding-bias groups available.",
  cols = c("trial_id", "digits_x", "n_values", "bias_up", "bias_down", "abs_bias_gap", "anomaly_flag"),
  n = 20
)
```

trial_id	digits_x	n_val- ues	bias_up	bias_down	abs_bias_gap	anomaly_flag
lung-time_c01_s41591_025_04181	0	2	0	0	0	FALSE

9 Statcheck Results

```
if (nrow(statcheck_raw) > 0) {
  statcheck_ranked <- statcheck_raw[order(statcheck_raw$error, decreasing = TRUE), ]
  knitr::kable(
    compact_df(
      statcheck_ranked,
      cols = c(
```

```

      "source", "test_type", "df1", "df2", "test_value",
      "reported_p", "computed_p", "error", "decision_error"
    ),
    n = 20,
    char_width = 36
  ),
  digits = 4
)
} else {
  cat("No statcheck output (package unavailable or no parseable report text).")
}

```

No statcheck output (package unavailable or no parseable report text).

10 Standardized Output Snapshot

```

render_table(
  standardized,
  "No standardized rows available.",
  cols = c(
    "trial_id", "method", "metric", "value_numeric",
    "p_value", "anomaly_flag", "severity", "details"
  ),
  n = 20,
  digits = 4,
  char_width = 40
)

```

trial_id	method	metric	value_numeric	p_value	anomaly_flag	severity	details
lung-time_c01_s41591ing025co04181	round-sistency	abs_percent_delta	0.0238	NA	FALSE	low	reported_percent=90.5; computed_perce...
lung-time_c01_s41591ing025co04181	round-sistency	abs_percent_delta	0.0238	NA	FALSE	low	reported_percent=90.5; computed_perce...

trial_id	method	metric	value_nu- meric	p_valu	anomaly	sever- ity	flag	details
lung- time_c01_s41591ing025co04181	round- sistency	abs_per- cent_delta	0.0238	NA	FALSE	low		reported_per- cent=9.5; computed_perce...
lung- time_c01_s41591ing025co04181	round- sistency	abs_per- cent_delta	0.0238	NA	FALSE	low		reported_per- cent=9.5; computed_perce...
lung- time_c01_s41591ing025co04181	round- sistency	abs_per- cent_delta	0.0000	NA	FALSE	low		reported_per- cent=80; computed_per- cent=80
lung- time_c01_s41591ing025co04181	round- sistency	abs_per- cent_delta	0.0429	NA	FALSE	low		reported_per- cent=82.9; computed_perce...
lung- time_c01_s41591ing025co04181	round- sistency	abs_per- cent_delta	0.0000	NA	FALSE	low		reported_per- cent=20; computed_per- cent=20
lung- time_c01_s41591ing025co04181	round- sistency	abs_per- cent_delta	0.0429	NA	FALSE	low		reported_per- cent=17.1; computed_perce...
lung- time_c01_s41591ing025co04181	round- sistency	abs_per- cent_delta	0.0095	NA	FALSE	low		reported_per- cent=36.2; computed_perce...
lung- time_c01_s41591ing025co04181	round- sistency	abs_per- cent_delta	0.0238	NA	FALSE	low		reported_per- cent=30.5; computed_perce...
lung- time_c01_s41591ing025co04181	round- sistency	abs_per- cent_delta	0.0095	NA	FALSE	low		reported_per- cent=63.8; computed_perce...
lung- time_c01_s41591ing025co04181	round- sistency	abs_per- cent_delta	0.0238	NA	FALSE	low		reported_per- cent=69.5; computed_perce...
lung- time_c01_s41591ing025co04181	round- sistency	abs_per- cent_delta	0.0429	NA	FALSE	low		reported_per- cent=57.1; computed_perce...
lung- time_c01_s41591ing025co04181	round- sistency	abs_per- cent_delta	0.0333	NA	FALSE	low		reported_per- cent=53.3; computed_perce...

trial_id	method	metric	value_nu- meric	p_valu- e	anomaly flag	sever- ity	details
lung- time_c01_s41591ing025co04181	round- sistency	abs_per- cent_delta	0.0429	NA	FALSE	low	reported_per- cent=42.9; computed_perce...
lung- time_c01_s41591ing025co04181	round- sistency	abs_per- cent_delta	0.0333	NA	FALSE	low	reported_per- cent=46.7; computed_perce...
lung- time_c01_s41591ing025co04181	round- sistency	abs_per- cent_delta	0.0429	NA	FALSE	low	reported_per- cent=22.9; computed_perce...
lung- time_c01_s41591ing025co04181	round- sistency	abs_per- cent_delta	0.0095	NA	FALSE	low	reported_per- cent=16.2; computed_perce...
lung- time_c01_s41591ing025co04181	round- sistency	abs_per- cent_delta	0.0429	NA	FALSE	low	reported_per- cent=77.1; computed_perce...
lung- time_c01_s41591ing025co04181	round- sistency	abs_per- cent_delta	0.0095	NA	FALSE	low	reported_per- cent=83.8; computed_perce...

11 Largest Rounding Deltas

```
ranked <- row_results[order(-row_results$abs_percent_delta, na.last = TRUE), ]
knitr::kable(head(ranked[, c(
  "variable", "level", "group", "reported_percent", "computed_percent", "abs_percent_delta"
)], 15), digits = 4)
```

	variable	level	group	reported_per- cent	com- puted_per- cent	abs_per- cent_delta
22	Brain metastasis	Yes	late_tod	14.3	15.2381	0.9381
23	Brain metastasis	No	early_tod	86.7	87.6190	0.9190
32	PD-L1 TPS	1-49%	late_tod	27.5	27.6190	0.1190
52	LIPI score	Medium risk	late_tod	35.3	35.2381	0.0619

	variable	level	group	reported_percent	computed_percent	abs_percent_delta
29	PD-L1 TPS	<1%	early_tod	39.0	39.0476	0.0476
42	Tumor burden	80 mm	late_tod	41.0	40.9524	0.0476
44	Tumor burden	<80 mm	late_tod	59.0	59.0476	0.0476
34	PD-L1 TPS	50%	late_tod	21.0	20.9524	0.0476
6	Smoking history	Smoker	late_tod	82.9	82.8571	0.0429
13	Histology	Squamous	early_tod	57.1	57.1429	0.0429
15	Histology	Adenocarcinoma	early_tod	42.9	42.8571	0.0429
19	Stage	IV	early_tod	77.1	77.1429	0.0429
37	ICIs	Sintilimab	early_tod	77.1	77.1429	0.0429
38	ICIs	Sintilimab	late_tod	77.1	77.1429	0.0429
8	Smoking history	Never smoked	late_tod	17.1	17.1429	0.0429

12 Diagnostics

```
hist(
  row_results$abs_percent_delta,
  breaks = 10,
  col = "gray80",
  border = "white",
  main = "Absolute Percent Delta Distribution",
  xlab = "|reported_percent - computed_percent|"
)
abline(v = 0.2, col = "red", lwd = 2, lty = 2)
```

Absolute Percent Delta Distribution

